

BOULMER, United KingdomWMO: **032400**Lat: **55.4208N**Lon: **1.6014W**Elev: **70**StdP: **14.66**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.8	30.8	21.5	16.2	32.7	23.6	17.9	34.1	40.5	47.9	35.3	46.0	9.0	260	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	69.9	61.4	67.5	60.0	65.6	59.0	63.4	67.7	61.9	65.7	60.5	64.0	12.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.6	82.0	65.3	60.2	78.0	63.8	58.8	74.1	62.5	28.8	67.8	27.7	65.7	26.8	64.2	71.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.6	27.0	23.7	24.4	75.5	3.7	1.7	21.8	76.7	19.6	77.7	17.5	78.7	14.8	79.9		
(5)			23.0	66.9	3.5	1.8	20.5	68.2	18.5	69.3	16.5	70.3	14.0	71.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.6	40.4	40.8	42.6	45.5	49.8	54.7	58.5	58.6	55.4	50.8	45.0	41.0		
	DBStd	7.92	4.46	4.69	4.64	4.47	4.30	3.89	3.56	3.35	3.62	4.23	4.68	5.40		
	HDD50	1481	297	259	232	148	56	4	0	0	3	41	160	281		
	HDD65	5974	761	677	693	585	470	310	203	201	288	441	601	744		
	CDD50	987	1	2	4	13	51	145	264	266	165	65	9	2		
	CDD65	4	0	0	0	0	0	1	2	1	0	0	0	0		
	CDH74	8	0	0	0	0	0	2	4	2	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	11.3	13.8	13.7	12.7	10.9	9.9	9.4	8.5	9.0	10.1	11.6	12.4	13.2		
	PrecAvg	28.5	2.4	1.9	1.7	2.1	1.9	2.4	2.6	2.2	2.2	2.8	3.2	2.6		
	PrecMax	40.6	6.3	4.6	3.9	6.1	3.9	6.4	6.3	6.0	5.3	6.7	7.5	5.5		
	PrecMin	21.3	0.7	0.2	0.4	0.3	0.4	0.6	0.4	0.4	0.6	0.7	1.0	1.0		
	PrecStd	4.9	1.2	1.1	1.0	1.7	0.9	1.5	1.4	1.2	1.3	1.2	1.4	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	55.5	58.5	62.5	66.1	71.2	73.6	72.9	71.0	64.5	59.3	56.6		
		MCWB	50.7	51.1	51.7	54.8	57.0	61.4	63.6	63.2	62.1	58.6	55.5	53.0		
	2%	DB	51.9	52.9	55.0	58.3	62.4	66.8	69.7	69.1	66.5	61.2	55.9	53.3		
		MCWB	48.4	49.0	49.3	51.4	54.9	58.5	61.6	61.3	59.7	56.4	52.3	50.1		
	5%	DB	49.7	50.7	52.5	55.6	59.6	63.9	67.4	66.9	63.8	59.0	53.9	51.4		
		MCWB	46.3	46.8	47.5	49.3	53.1	56.6	60.1	60.1	57.8	54.8	50.6	48.4		
	10%	DB	47.4	48.5	50.3	53.0	57.1	61.7	65.3	64.8	61.6	57.0	52.1	49.0		
		MCWB	44.4	44.8	45.8	47.8	51.7	55.7	59.2	59.2	56.6	53.4	49.5	46.3		
	0.4%	WB	51.1	51.9	52.6	55.9	59.0	63.0	65.4	65.4	63.7	59.7	55.9	53.3		
		MCDB	53.9	54.7	57.3	61.0	64.1	68.9	70.9	69.7	69.0	62.9	58.5	56.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	48.8	49.3	50.1	52.6	56.4	60.0	63.3	63.4	60.8	57.4	53.3	50.7		
		MCDB	51.6	52.5	54.0	57.3	60.8	64.7	67.8	67.1	65.1	60.1	55.3	53.1		
	5%	WB	46.7	47.3	48.2	50.3	54.2	58.2	61.8	61.9	59.1	55.7	51.5	48.7		
		MCDB	49.2	50.4	52.0	54.1	58.1	62.3	65.8	65.1	62.1	58.0	53.1	51.0		
	10%	WB	44.9	45.2	46.3	48.4	52.5	56.6	60.3	60.6	57.6	54.3	49.9	46.6		
		MCDB	47.0	48.0	49.8	52.0	56.0	60.4	64.0	63.5	60.4	56.4	51.6	48.8		
	5% DB	MDBR	8.1	9.2	10.4	10.6	11.2	11.0	11.7	11.6	11.3	9.9	9.1	8.3		
		MCDBR	10.9	11.4	13.7	15.0	15.1	15.3	15.0	14.4	14.5	12.1	11.0	10.9		
		MCWBR	9.8	9.3	9.7	9.8	9.3	9.2	8.9	8.6	9.4	8.9	9.4	9.9		
		MCWBR	10.5	11.1	13.2	14.1	13.7	13.6	13.5	12.3	13.0	11.1	9.9	10.5		
	5% WB	MCWBR	9.8	9.5	10.2	9.9	9.2	9.1	9.0	8.8	9.5	9.1	8.9	9.8		
		taub	0.317	0.334	0.365	0.389	0.385	0.387	0.389	0.379	0.360	0.344	0.318	0.300		
		taud	2.307	2.382	2.320	2.284	2.348	2.368	2.370	2.409	2.463	2.457	2.365	2.271		
		Ebn at Noon	190	235	253	264	272	272	269	265	254	229	189	166		
		Edh at Noon	17	23	31	36	36	36	35	32	27	21	16	14		